

SIMATS ENGINEERING

Assessment Tool 3 — Resolution Algorithm: Worked Solutions
Course: Artificial Intelligence (CSA17) | CO3 | Annexure A

Question 1 – Rain and Wet Ground

Symbols: R = It rains, W = The ground is wet

1. Propositional logic: Premise 1: $R \rightarrow W$. Premise 2: R.
2. Convert to CNF: $R \rightarrow W$ becomes $\neg R \vee W$ (Clause A). R stays as R (Clause B).
3. Goal: W. Negate the goal: $\neg W$ (Clause C).
4. Resolution steps:
Step 1: Resolve Clause A ($\neg R \vee W$) with Clause C ($\neg W$) $\rightarrow$ resolve on W $\rightarrow \neg R$ (Clause D).
Step 2: Resolve Clause D ($\neg R$) with Clause B (R) $\rightarrow$ resolve on R $\rightarrow$ Empty Clause (■).

Conclusion: Empty clause (■) is derived, so the goal 'the ground is wet' is **PROVED**.

Question 2 – Student Assignment Submission

Symbols: S = A student submits the assignment, M = The student receives internal marks

1. Propositional logic: Premise 1: $S \rightarrow M$. Premise 2: S (Rahul submitted).
2. Convert to CNF: $S \rightarrow M$ becomes $\neg S \vee M$ (Clause A). S stays as S (Clause B).
3. Goal: M. Negate the goal: $\neg M$ (Clause C).
4. Resolution steps:
Step 1: Resolve Clause A ($\neg S \vee M$) with Clause C ($\neg M$) $\rightarrow$ resolve on M $\rightarrow \neg S$ (Clause D).
Step 2: Resolve Clause D ($\neg S$) with Clause B (S) $\rightarrow$ resolve on S $\rightarrow$ Empty Clause (■).

Conclusion: Empty clause (■) is derived, so the goal 'Rahul receives internal marks' is **PROVED**.

Question 3 – Library Membership

Symbols: L = A person is a library member, B = The person can borrow books

1. Propositional logic: Premise 1: $L \rightarrow B$. Premise 2: L (Priya is a member).
2. Convert to CNF: $L \rightarrow B$ becomes $\neg L \vee B$ (Clause A). L stays as L (Clause B).
3. Goal: B. Negate the goal: $\neg B$ (Clause C).
4. Resolution steps:
Step 1: Resolve Clause A ($\neg L \vee B$) with Clause C ($\neg B$) $\rightarrow$ resolve on B $\rightarrow \neg L$ (Clause D).
Step 2: Resolve Clause D ($\neg L$) with Clause B (L) $\rightarrow$ resolve on L $\rightarrow$ Empty Clause (■).
No new clauses can be generated after this — resolution terminates.

Conclusion: Empty clause (■) is derived, so the goal 'Priya can borrow books' is **PROVED**.

Question 4 – Placement Eligibility

Symbols: A = A student clears the aptitude test, P = The student is eligible for placement

1. Propositional logic: Premise 1: $A \rightarrow P$. Premise 2: A (Arun cleared the test).
2. CNF clauses: $\neg A \vee P$ (Clause 1). A (Clause 2).

3. Goal: P. Negate the goal: $\neg P$ (Clause 3).

4. Resolution steps:

Step 1: Resolve Clause 1 ($\neg A \vee P$) with Clause 3 ($\neg P$) $\rightarrow$ resolve on P $\rightarrow \neg A$ (Clause 4).

Step 2: Resolve Clause 4 ($\neg A$) with Clause 2 (A) $\rightarrow$ resolve on A $\rightarrow$ Empty Clause (■).

5. Resolution tree: $[\neg A \vee P]$ and $[\neg P]$ join at P $\rightarrow$ produce $[\neg A]$; $[\neg A]$ and $[A]$ join at A $\rightarrow$ produce [■]. The tree has two resolution steps ending in the empty clause.

Conclusion: Empty clause (■) is derived, so the goal 'Arun is eligible for placement' is PROVED.

Question 5 – Access Control System

Symbols: C = User enters the correct password, U = User is authenticated, G = User is granted access

1. Propositional logic: Premise 1: $C \rightarrow U$. Premise 2: $U \rightarrow G$. Premise 3: C.

2. CNF clauses: $\neg C \vee U$ (Clause 1). $\neg U \vee G$ (Clause 2). C (Clause 3).

3. Goal: G. Negate the goal: $\neg G$ (Clause 4).

4. Resolution steps:

Step 1: Resolve Clause 2 ($\neg U \vee G$) with Clause 4 ($\neg G$) $\rightarrow$ resolve on G $\rightarrow \neg U$ (Clause 5).

Step 2: Resolve Clause 5 ($\neg U$) with Clause 1 ($\neg C \vee U$) $\rightarrow$ resolve on U $\rightarrow \neg C$ (Clause 6).

Step 3: Resolve Clause 6 ($\neg C$) with Clause 3 (C) $\rightarrow$ resolve on C $\rightarrow$ Empty Clause (■).

Explanation: Each step eliminates one literal and its negation, chaining the password $\rightarrow$ authentication $\rightarrow$ access implications until a contradiction (empty clause) is reached.

Conclusion: Empty clause (■) is derived, so the goal 'the user is granted access' is PROVED.